Digital applications and extended-answer technique

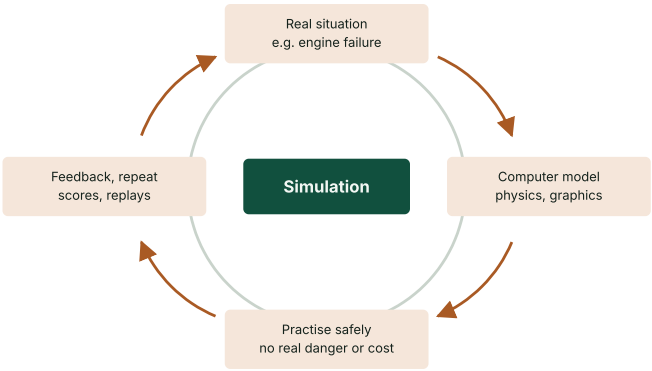
L24 FACT FILE

Specification: students should be able to

- Describe the main features of gaming applications, simulations and mobile phone applications
- Describe how gaming applications, simulations and mobile phone applications can be used to support education and training, social interactions and work practices
- Describe how virtual reality is used to create realistic simulations for training
- Evaluate the impact of online banking on customers and on banks
- Evaluate the impact of online training on employees and employers
- Evaluate the impact of e-commerce on customers and on businesses

CCEA fact file: Digital Applications

Key words: gaming application, cloud gaming, leaderboard, gamification, immediate feedback, simulation, computer model, virtual reality, mobile app, app store, online banking, two-factor authentication, phishing, online training, automatic tracking, e-commerce, evaluate, quality of written communication



How a simulation supports training: model, practise safely, get feedback, repeat

Digital applications in everyday life

Digital applications are programs that help people learn, train, stay in touch, work, bank and shop. Gaming applications use rules, levels, points, leaderboards and immediate feedback to motivate players, and online multiplayer lets people play and talk together in real time. Simulations use a computer model, built from data and mathematical rules, to copy how a real object or situation behaves, so people can test designs or practise dangerous tasks safely. Virtual reality takes this further by placing the user inside a 3D world through a headset. Mobile phone apps are always with the user and use features such as notifications, GPS and the camera to support learning, communication and work on the move. Online banking, online training and e-commerce have changed how we manage money, gain skills and buy goods. Each brings clear benefits, such as 24/7 access and convenience, but also drawbacks, such as fraud risk, the need for internet access and digital skills, and less face-to-face contact. In the exam you must be able to describe the features and uses of the first three and evaluate the impact of the last three on both users and organisations.

Games, simulations and apps at work and school

Games suit training because they are built in levels: each level brings a new challenge and you cannot skip ahead, so skills are built step by step. Immediate feedback shows the learner what went well and what to try next, and role play lets them make decisions and reflect on the results without real consequences. Leaderboards and badges, often called gamification when they are added to non-game tasks, encourage people to keep improving. Simulations are used where real practice would be dangerous, expensive or rare. A trainee pilot can face engine failure or a thunderstorm in a flight simulator, with every action recorded for feedback, and no fuel is used and no aircraft is taken out of service. Surgeons can rehearse an operation on a 3D virtual patient made from scan data. Mobile apps let teachers share video, audio and e-books, let pupils reach their work through the cloud at home and school, let managers message groups of staff, and let schools tell parents about events. Good apps load quickly, keep personal data secure, can be personalised and are kept up to date by their developers.

Evaluating online banking and online training

Online banking lets customers check balances, view statements, pay bills, transfer money and set up standing orders and direct debits at any time, even from abroad, and Faster Payments moves money almost instantly. However, customers need a device, an internet connection and the skills to use it, so some older or less confident people are excluded. Fraud is a real worry: phishing emails and fake texts try to steal login details. Some customers miss face-to-face help. For banks, online services cut costs because fewer branches and staff are needed, and e-statements save paper and postage, but banks must protect huge amounts of personal data with complex, constantly updated security and must employ specialist staff. Online training lets employees learn at their own pace and at a convenient time, saves travel, lets staff in different places train at once and records progress automatically. Multimedia and VR simulations make it engaging and realistic. On the other hand, support may be limited to FAQs or email, learners can feel isolated, it needs self-motivation, deadlines are hard to manage and a poor connection can stop learning altogether.

Evaluating e-commerce

E-commerce means buying and selling goods and services over the internet. For customers it is convenient: they can shop at any time without travelling, which especially helps people with disabilities, parents of young children and those living far from town centres. They can read reviews, compare prices, choose from a far wider range than a local shop stocks and pre-order popular items. For businesses, there is no need for expensive high-street premises or checkout staff, so savings can be passed on as lower prices, and a website can reach customers worldwide. The drawbacks are also real. Customers cannot see or try goods before buying, so items may not match the photos and must be posted back. They wait for delivery and often pay extra for it, and fake websites may be set up to steal card details. Secure payment services and checking for HTTPS reduce this risk. In the UK, the Consumer Contracts Regulations 2013 usually give online shoppers 14 days after receiving goods to cancel the order. High-street shops may close as more people shop online, affecting town centres and jobs.

T Key terms from the mark schemes

E-commerce
The buying and selling of goods and services over the internet.
Credited: Online shopping; electronic commerce

Online banking
Using a bank's secure website or app to manage an account, for example viewing statements, paying bills and transferring money.
Credited: Internet banking

Online training
Learning delivered over the internet, often with multimedia and automatic progress tracking, that learners can complete at their own pace.
Credited: Web-based training; e-learning

Gaming application
An electronic game played on a console, computer or mobile device, often online with other players.

Gamification
Adding game features such as points, badges and leaderboards to a non-game activity to increase motivation.

Simulation
A computer model, built from data and mathematical rules, that copies how a real object or situation behaves so it can be tested or practised safely.
Credited: Computer model

Virtual reality
A computer-generated 3D environment that the user enters using a headset and sometimes gloves or controllers, stimulating senses such as sight, hearing and touch.
Credited: VR

Mobile app
A software application designed to run on a smartphone or tablet, usually downloaded from an app store.
Credited: Mobile phone application

Phishing
Fake emails, texts or websites that pretend to be from a trusted organisation in order to trick people into giving personal or login details.

Two-factor authentication
Checking a user's identity using two different methods, such as a password plus a code sent to their phone.
Credited: 2FA

GUI
Graphical user interface: lets users interact using windows, icons, menus and a pointer, or touch on a mobile app.
Credited: WIMP interface

Quality of written communication
Credit in extended answers for organised writing, correct spelling and grammar, and accurate use of specialist terms.
Credited: QWC

Exam focus

This topic is almost always tested through 6-mark extended questions that also assess quality of written communication: advantages of online training (2019 Q9), disadvantages of online shopping for customers (2022 Q6) and disadvantages of online banking to the customer (2025 Q12). Shorter questions have asked for drawbacks of online learning (2023 Q11(b)), risks of gaming (2024 Q9) and matching key terms such as e-commerce (2021 Q11). Answers are marked in levels: roughly Level 1 (1-2 marks) for a few basic points, Level 2 (3-4) for some developed points and Level 3 (5-6) for several well-developed, accurate points in an organised answer using specialist terms. Read the question carefully: if it asks about the customer, points about the bank or business earn nothing, and if it asks for disadvantages, advantages are not credited. Develop every point with a reason or consequence. Words such as 'faster', 'easier' or 'cheaper' on their own are not credited, so explain what is faster and why it matters. Spell key terms correctly and write in paragraphs.

Unit 1 2025 Q12	[6]	Disadvantages of online banking to the customer (QWC)
Unit 1 2024 Q9	[7]	Input devices, health risks, gaming and internet risks
Unit 1 2023 Q11(b)	[2]	Drawbacks of online learning
Unit 1 2022 Q6	[6]	Disadvantages of online shopping for customers (QWC)
Unit 1 2021 Q11	[8]	Key term matching

? Check yourself

1. State what is meant by e-commerce. [1]
2. State two features of gaming applications that help to motivate learners. [2]
3. Describe how gaming applications support social interaction. [2]

Answers: 1. Buying and selling goods or services over the internet (1). 2. Any two from: leaderboards (1) / points or scores (1) / badges or rewards (1) / levels of increasing difficulty (1) / immediate feedback (1) / multimedia such as animation and sound (1). 3. Any one developed point, e.g. online multiplayer lets players compete or co-operate with others anywhere in the world (1) and talk in real time using headsets and microphones or text chat (1). Accept joining online gaming groups/communities and collaborating to solve problems.